Amar Kushwaha

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Education

Texas Tech University **Dec 2027**
Bachelor of Science in Computer Science, Minor in Mathematics, Honors College *GPA: 4.0*

Technical Skills

Languages: Python, JavaScript/TypeScript, SQL, C, Rust, HTML/CSS
Web/Backend: React, Next.js, Node.js, FastAPI, REST APIs, PostgreSQL, Docker, Linux
AI/ML: PyTorch, TensorFlow, scikit-learn, LangChain, Gemini API, FAISS, Numba JIT, MNE-Python
Engineering: Git, GitHub Actions, CI/CD automation, benchmarking, and performance tuning

Experience

Applied Scientist Intern – Agentic AI **May 25, 2026 – Aug. 7, 2026**
Tempora Labs *Remote | TX*

- Ship user-facing agent workflows end-to-end—planning, tool-augmented reasoning, multi-agent orchestration, and human-in-the-loop review—with attention to reliability and clear failure modes.
- Implement retrieval-augmented and tool-calling LLM systems for automated digital-asset investment flows: structured function calls, ReAct-style reasoning, planner–executor decomposition, reflection loops, and coordinated specialist agents.
- Drive model and serving quality via fine-tuning, prompt engineering, systematic offline/online **evaluations**, and inference optimization (attention, positional encodings, KV caching); integrate external market and on-chain data behind guarded interfaces.
- Partner with engineering, research, and product in agile iteration to land production features; translate architecture and tradeoffs for internal teams and enterprise customers through specs, deep-dives, and demos.

Undergraduate Research Scholar **March 2025 – Present**
ECE Department | TTU Honors College | TrUE Program *Lubbock, TX*

- Build and simulate large-scale brain network models (Epileptor 6D) to study seizure propagation across **76–998 region** connectomes with reproducible computational pipelines.
- Accelerate numerical experiments with **Numba JIT** (∼1000× speedup) and realistic transmission-delay models for closed-loop control prototype methods.
- Integrate Human Connectome Project tractography with interactive 3D visualization tooling to support data-driven surgical planning workflows.
- Co-author peer-reviewed work in **Frontiers in Network Physiology – Fractal Physiology**.

Projects

Universal Sentinel | *Python, FastAPI, React, REST, Solana/EVM, Gemini AI* **GitHub**

- Built a full-stack incident response workflow: ingestion from **GDACS, NASA, NOAA**; FastAPI services and a React UI orchestrating verification, treasury logic, and payouts.
- Designed an auditable decision pipeline (observe → LLM judge → on-chain vault) with NGO checks, immutable logs, and traceable fund movements.
- Implemented deterministic routing (region/disaster matching) alongside ML judgment to reduce operational risk and simplify monitoring.

git0x-cli | *Node.js, JavaScript, npm, SARIF, GitHub Actions* **GitHub**

- Delivered a Node.js CLI secret scanner (**70+** patterns) covering AI keys, cloud credentials, databases, and payment tokens—security posture aligned with enterprise CI expectations.
- Integrated SARIF + JSON reporting, GitHub Actions, and pre-commit hooks to block leaks early with baseline/allowlist controls for manageable noise.

Achievements

Hook 'Em Hacks: Winner, Best Use of Solana – *Oath*: pre-commitment and enforcement protocol for AI agents
ETHDenver 2026: Dual Track Winner with Kayden – Nouns Builder (Governance UX & Community Tooling) + ADI Foundation Track
TAMU Datathon 2025: Winner, HackMLH – enterprise document workflow with Gemini AI, structured extraction, and audit-friendly Solana logging
HackWesTX VI: Dual Winner – "Best Use of Gemini API" + 1st Place "Best Use of Web3 Solana" (production-minded full-stack delivery)